E.C. - Math challenge #1
ENGR& 204 #17385

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1 Problem 1.

A semicircle is inscribed in a quarter circle, and one of its diameter endpoints is a distance 1 from the center of the quarter circle, drawn below. What fraction of the quarter circle's area is contained in the semicircle? (i.e. what fraction of the quarter circle is shaded?)

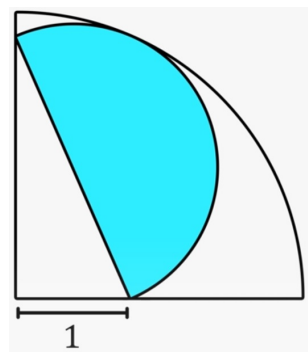
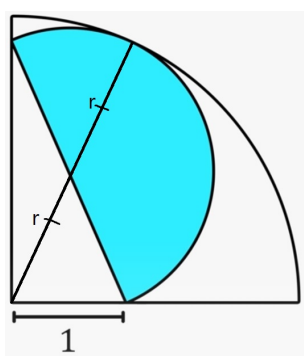


Figure 1: Given diagram.

SOLUTION: Something to notice is that when drawing from center of a circle to their end results in a tangent line to the curve. This means when we draw a line from the bigger, outer circle the tangent point, it will also hit the center of the inner circle. With this, we can calculate the areas symbolically. Since, we're only concerned about the ratio or fraction, we don't actually need the exact value.



$$A_{\text{big quarter}} = \frac{\pi(\cancel{2}r)^2}{\cancel{4}} = \pi r^2$$

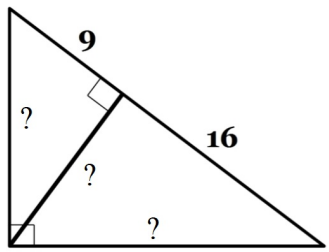
$$A_{\text{small semi}} = \frac{\pi r^2}{2}$$

Clearly, we can see that half of the big quarter circle is enclosed by the small semi circle.

$$\boxed{\frac{A_{\text{small semi}}}{A_{\text{big quarter}}} = \frac{1}{2}}$$

2 Problem 2.

A right triangle has a line drawn from the vertex of the right angle to the hypotenuse, making two more right angles – this is called the perpendicular. This perpendicular divides the triangle into two segments that have lengths of 9 and 16 respectively. Find the length of the perpendicular, and the two legs of the triangle as shown below.



SOLUTION: There are many ways to solving this and the easiest is probably the usage of similair triangles in combination with some Pythaogrean theorem. Look at how the altitude splits the right angle into two complementary angles. From this relationship, we also know that the other smaller triangle will share the same angles and therefore the same ratios hence similiar triangles. Using this, we can find the altitude length.

Let a be the altitude side $\overline{BE}$.

$$\begin{aligned} \frac{9}{a} &= \frac{a}{16} \\ (9)(16) &= a^2 \\ a &= 12, \text{ lengths are only positive.} \end{aligned}$$

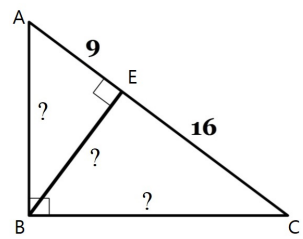


Figure 2: $\triangle ABC \sim \triangle AEB \sim \triangle BEC$

Now we can use Pythaogrean Theorem on both sides.

$$\begin{aligned} \overline{AB}^2 &= 9^2 + 12^2 \\ \overline{AB} &= 15 \\ \overline{BC}^2 &= 16^2 + 12^2 \\ \overline{BC} &= 20 \end{aligned}$$

In whole, these are all of our lengths. $\boxed{\overline{BE} = 12,\ \overline{AB} = 15,\ \overline{BC} = 20}$

2.1 Extra Methods

Just thought it was interesting with the variety of methods. My actual initial thought was to create a system of equation and spam Pythaogrean Theorem.

$$\begin{aligned} \overline{BE}^2 + 9^2 &= \overline{AB}^2 \\ \overline{BE}^2 + 16^2 &= \overline{BC}^2 \\ \overline{AB}^2 + \overline{BC}^2 &= 25^2 \end{aligned}$$

Solving this isn't actually that bad because we can substitue all of the squared varables with just another variable and then unsquare it when we are done.

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \overline{AB}^2 \\ \overline{BC}^2 \\ \overline{BE}^2 \end{bmatrix} = \begin{bmatrix} -81 \\ -256 \\ 625 \end{bmatrix}$$

$$\begin{bmatrix} \overline{AB}^2 \\ \overline{BC}^2 \\ \overline{BE}^2 \end{bmatrix} = \begin{bmatrix} 225 \\ 400 \\ 114 \end{bmatrix}$$

$$\begin{bmatrix} \overline{AB} \\ \overline{BC} \\ \overline{BE} \end{bmatrix} = \begin{bmatrix} 15 \\ 20 \\ 12 \end{bmatrix}$$

Another method quite literally is the exact usage of the Geometric mean theorem in which we can solve for the altitude.

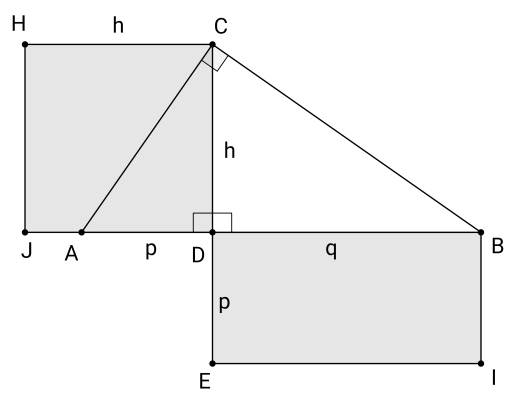


Figure 3: From Wikipedia $h = \sqrt{pq}$.

$$h = \sqrt{16 \cdot 9}$$
$$\implies h = 12 = \overline{BE}$$

From which, we can simply solve for both side legs in the original method.